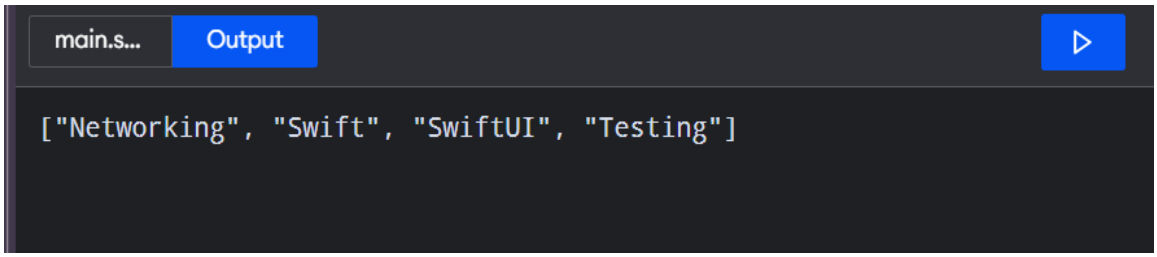
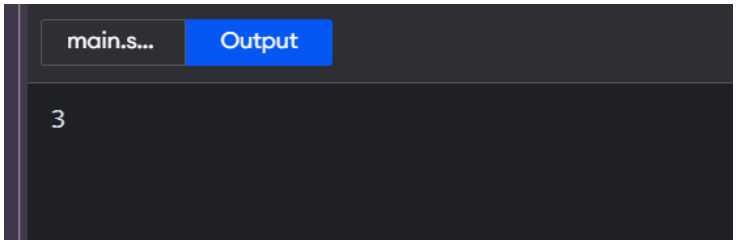
Day 3:
1 i)

```
var subjects = ["Swift", "UIKit", "SwiftUI"]
subjects.append("Networking")
subjects.insert("Testing", at:1)
subjects.removeAll {$0 == "UIKit"}
subjects.sort()
print(subjects)
```



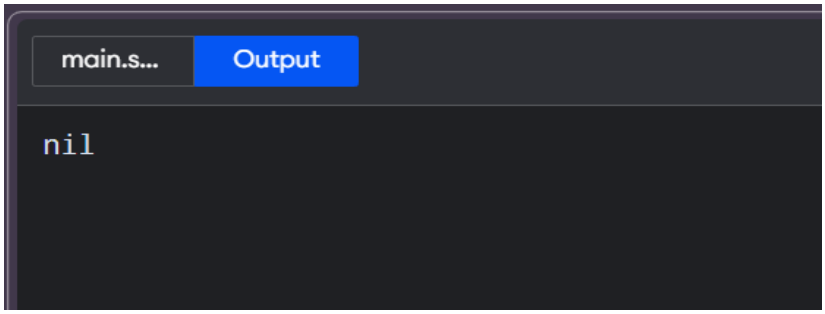
ii)

```
var registered: Set<String> = ["Arun", "Priya"]
registered.insert("Kumar")
registered.insert("Kumar")
print(registered.count)
```



iii)

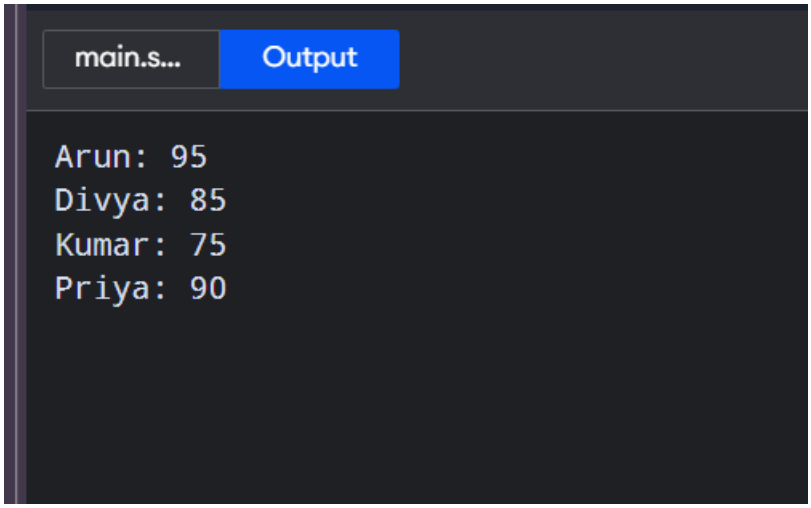
```
var marks: [String: Int] = [
    "Arun": 80,
    "Priya": 90,
    "Kumar": 75
]
marks["Divya"] = 85
marks["Arun"] = 95
print(marks["Ravi"] as Any)
```



iv)

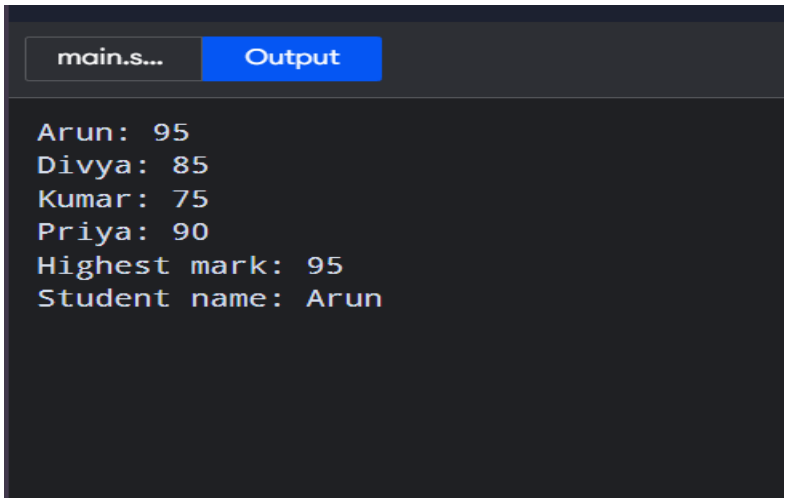
```
var marks: [String: Int] = [
    "Arun": 80,
    "Priya": 90,
    "Kumar": 75
]
marks["Divya"] = 85
marks["Arun"] = 95
```

```
for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}
```



v)let highest = marks.max(by: { $0.value < $1.value })!

```
print("Highest mark: \(highest.value)")
print("Student name: \(highest.key)")
```



Task 2:

```
let sentence = "swift is fun and swift is fast"

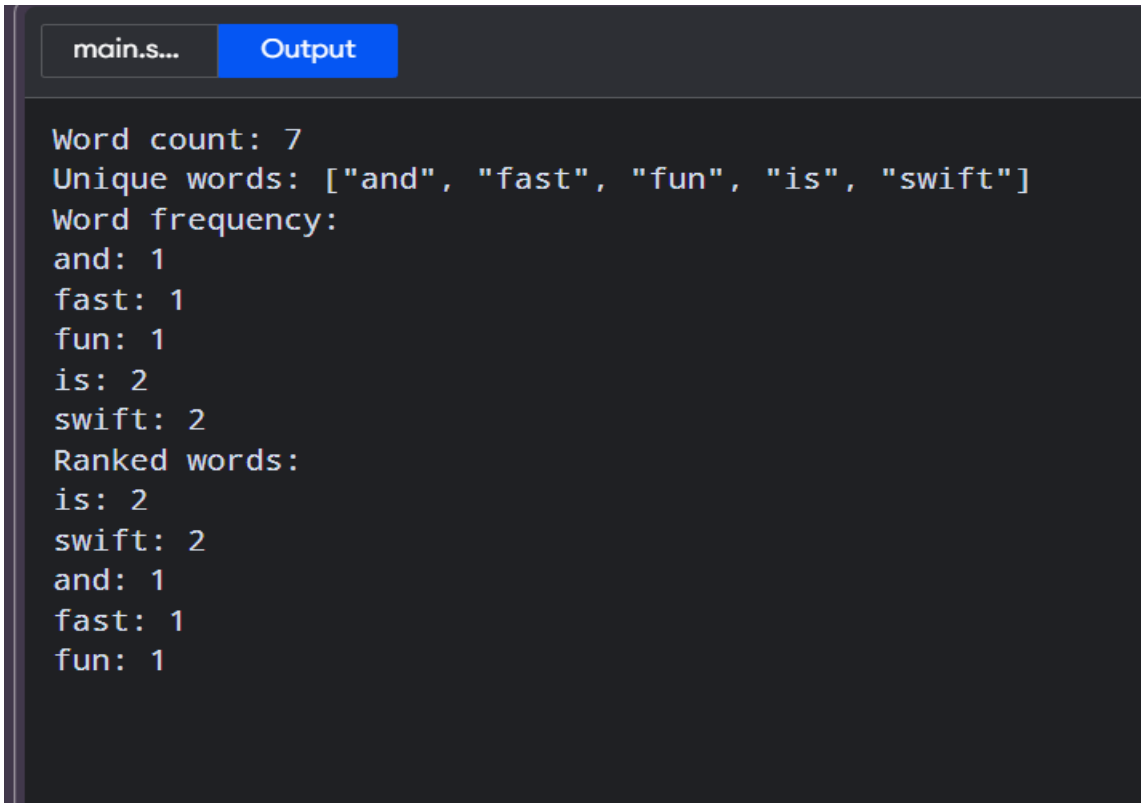
let words = sentence.split(separator: " ").map { String($0) }
print("Word count:", words.count)
let uniqueWords = Set(words)
print("Unique words:", uniqueWords.sorted())
var frequency: [String: Int] = [:]

for word in words {
    frequency[word, default: 0] += 1
}

print("Word frequency:")
for word in frequency.keys.sorted() {
    print("\(word): \(frequency[word]!)")
}
```

```
let rankedWords = frequency.sorted {
    if $0.value != $1.value {
        return $0.value > $1.value
    }
    return $0.key < $1.key
}

print("Ranked words:")
for (word, count) in rankedWords {
    print("\(word): \(count)")
}
```

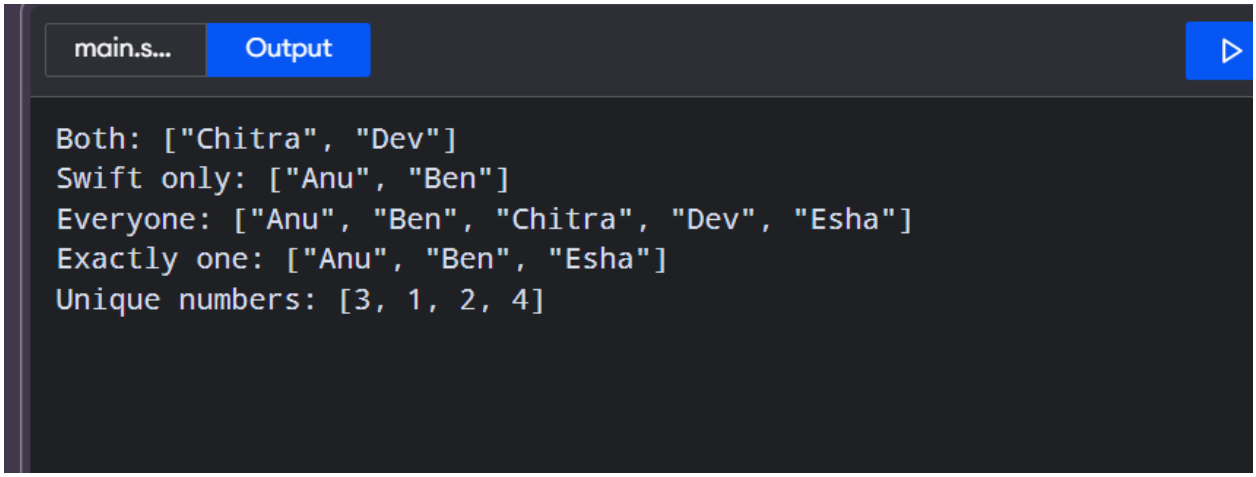


Task 3:

```
let swiftClub: Set<String> = ["Anu", "Ben", "Chitra", "Dev"]
let uiClub: Set<String> = ["Chitra", "Dev", "Esha"]
print("Both:", swiftClub.intersection(uiClub).sorted())
print("Swift only:", swiftClub.subtracting(uiClub).sorted())
print("Everyone:", swiftClub.union(uiClub).sorted())
print("Exactly one:", swiftClub.symmetricDifference(uiClub).sorted())
let numbers = [3, 1, 3, 2, 1, 4]
var uniqueNumbers: [Int] = []

for number in numbers {
    if !uniqueNumbers.contains(number) {
        uniqueNumbers.append(number)
    }
}

print("Unique numbers:", uniqueNumbers)
```


main.s...
Output
Both: ["Chitra", "Dev"]
Swift only: ["Anu", "Ben"]
Everyone: ["Anu", "Ben", "Chitra", "Dev", "Esha"]
Exactly one: ["Anu", "Ben", "Esha"]
Unique numbers: [3, 1, 2, 4]